Multivariable Calculus: Problem Set 6

Detailed Assignment-Style Solutions with Diagrams

Lecturer: Rajendra Bhatia TA: Debabrata Jana

Global conventions used throughout

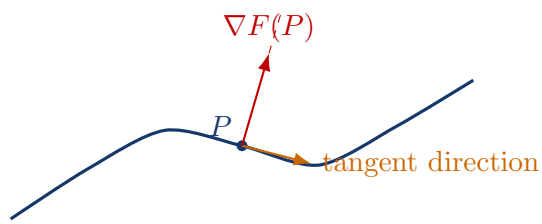
For a level curve $F(x, y) = c$, the gradient $\nabla F(x_0, y_0)$ is perpendicular to the tangent line at (x_0, y_0), provided $\nabla F(x_0, y_0) \neq 0$. Therefore the tangent line is

$$F_x(x_0, y_0)(x - x_0) + F_y(x_0, y_0)(y - y_0) = 0.$$

For a level surface $F(x, y, z) = c$, the gradient $\nabla F(x_0, y_0, z_0)$ is perpendicular to the tangent plane, so the tangent plane is

$$F_x(P)(x - x_0) + F_y(P)(y - y_0) + F_z(P)(z - z_0) = 0,$$

where $P = (x_0, y_0, z_0)$.



On a level curve, the gradient is normal to the curve.
The tangent line is therefore perpendicular to the gradient.

Problem 1: Level curves and tangent lines

Method. Write the curve as $F(x, y) = k$ or $F(x, y) = 0$. Compute ∇F. At the specified point $P = (x_0, y_0)$, use $\nabla F(P)$ as the normal vector to the tangent line.

1(i) $x^2 + y^2 = k^2$ at $(k, 0)$

Let

$$F(x, y) = x^2 + y^2.$$

The level curve is $F(x, y) = k^2$. Its gradient is

$$\nabla F(x, y) = (2x, 2y).$$

At $(k, 0)$,

$$\nabla F(k, 0) = (2k, 0).$$

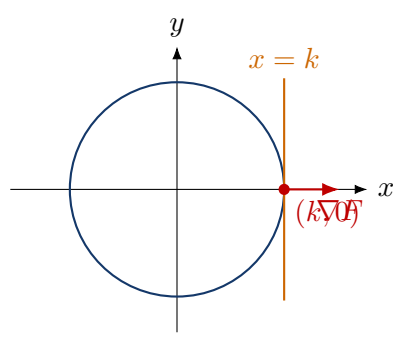
Thus the tangent line satisfies

$$2k(x - k) + 0(y - 0) = 0.$$

If $k \neq 0$, divide by $2k$:

$$x - k = 0.$$

Answer. $\boxed{x = k}$. If $k = 0$, the curve collapses to the single point $(0, 0)$, so there is no ordinary tangent line to a regular curve.



The radius is normal to the circle, so the tangent is vertical.

1(ii) $xy = k$ at $(1, k)$

Let

$$F(x, y) = xy.$$

Then

$$\nabla F(x, y) = (y, x).$$

At $(1, k)$,

$$\nabla F(1, k) = (k, 1).$$

Therefore the tangent line is

$$k(x-1) + 1(y-k) = 0.$$

Equivalently,

$$y - k = -k(x-1).$$

Answer. $\boxed{k(x-1) + y - k = 0}$, or $\boxed{y = -kx + 2k}$.

1(iii) $y - x^2 = k$ at $(1, k+1)$

Let

$$F(x, y) = y - x^2.$$

Then

$$\nabla F(x, y) = (-2x, 1).$$

At $(1, k+1)$,

$$\nabla F(1, k+1) = (-2, 1).$$

The tangent line is

$$-2(x-1) + 1(y - (k+1)) = 0.$$

So

$$y - k - 1 = 2(x-1),$$

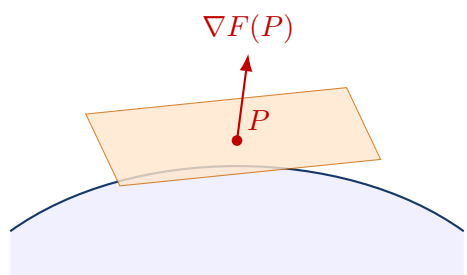
and therefore

$$y = 2x + k - 1.$$

Answer. $\boxed{y = 2x + k - 1}$.

Problem 2: Level surfaces and tangent planes

Method. Either convert the surface to a level surface $F(x, y, z) = 0$ and use ∇F as the normal vector, or, if the surface is a graph $z = f(x, y)$, use $z - z_0 = f_x(P)(x - x_0) + f_y(P)(y - y_0)$. Always first check that the given point lies on the surface.



For a level surface, the tangent plane is the plane perpendicular to $\nabla F(P)$.

2(i) $z = x^2 + y^2$ at $(1, 1, 2)$

Here $z = f(x, y) = x^2 + y^2$. First check the point:

$$1^2 + 1^2 = 2,$$

so $(1, 1, 2)$ lies on the surface. Compute

$$f_x(x, y) = 2x, \qquad f_y(x, y) = 2y.$$

At $(1, 1)$,

$$f_x(1, 1) = 2, \qquad f_y(1, 1) = 2.$$

Therefore

$$z - 2 = 2(x - 1) + 2(y - 1).$$

Simplifying,

$$z = 2x + 2y - 2.$$

Answer. $\boxed{z = 2x + 2y - 2}$.

2(ii) $y = x^2 - z^2$ at $(4, 7, 2)$

First check whether the point lies on the surface. The right-hand side at $(x, z) = (4, 2)$ is

$$4^2 - 2^2 = 16 - 4 = 12.$$

But the given y-coordinate is 7, not 12. Hence

$$7 \neq 12.$$

Answer. The point $\boxed{(4,7,2)}$ is not on the surface $y = x^2 - z^2$. Therefore the requested tangent plane at that point is not defined.

Remark. If the intended point was $(4,12,2)$, then $y - 12 = 8(x-4) - 4(z-2)$, i.e. $y = 8x - 4z - 12$.

2(iii) $xy + yz + zx = 5$ at $(1,2,1)$

Let

$$F(x,y,z) = xy + yz + zx - 5.$$

Check the point:

$$1 \cdot 2 + 2 \cdot 1 + 1 \cdot 1 = 2 + 2 + 1 = 5.$$

So the point lies on the surface. Compute

$$F_x = y + z, \qquad F_y = x + z, \qquad F_z = x + y.$$

At $(1,2,1)$,

$$\nabla F(1,2,1) = (3,2,3).$$

Thus the tangent plane is

$$3(x-1) + 2(y-2) + 3(z-1) = 0.$$

Simplifying,

$$3x + 2y + 3z = 10.$$

Answer. $\boxed{3x + 2y + 3z = 10}$.

2(iv) $\dfrac{x^2}{a^2} + \dfrac{y^2}{b^2} + \dfrac{z^2}{c^2} = 1$ at (x_0, y_0, z_0)

Let

$$F(x,y,z) = \frac{x^2}{a^2} + \frac{y^2}{b^2} + \frac{z^2}{c^2} - 1.$$

Then

$$\nabla F(x,y,z) = \left(\frac{2x}{a^2}, \frac{2y}{b^2}, \frac{2z}{c^2}\right).$$

At (x_0, y_0, z_0),

$$\nabla F(x_0,y_0,z_0) = \left(\frac{2x_0}{a^2}, \frac{2y_0}{b^2}, \frac{2z_0}{c^2}\right).$$

The tangent plane is

$$\frac{2x_0}{a^2}(x - x_0) + \frac{2y_0}{b^2}(y - y_0) + \frac{2z_0}{c^2}(z - z_0) = 0.$$

Dividing by 2,

$$\frac{x_0 x}{a^2} + \frac{y_0 y}{b^2} + \frac{z_0 z}{c^2} = \frac{x_0^2}{a^2} + \frac{y_0^2}{b^2} + \frac{z_0^2}{c^2}.$$

Since (x_0, y_0, z_0) lies on the ellipsoid, the right-hand side is 1. Hence

$$\frac{x_0 x}{a^2} + \frac{y_0 y}{b^2} + \frac{z_0 z}{c^2} = 1.$$

Answer. $\boxed{\frac{x_0 x}{a^2} + \frac{y_0 y}{b^2} + \frac{z_0 z}{c^2} = 1}$.

Problem 3: Chain rule

Method. When $f = f(x(t), y(t))$, use $\dfrac{df}{dt} = f_x \dfrac{dx}{dt} + f_y \dfrac{dy}{dt}$. When $f = f(x(t), y(t), z(t))$, use $\dfrac{df}{dt} = f_x x'(t) + f_y y'(t) + f_z z'(t)$. The computation is always: first compute partial derivatives, then compute derivatives of the inner functions, then substitute.

$$\boxed{t} \xrightarrow{\text{inner derivatives}} \boxed{(x(t), y(t), z(t))} \xrightarrow{\text{gradient}} \boxed{f(x, y, z)}$$

$$\frac{df}{dt} = \nabla f \cdot \frac{d}{dt}(x, y, z)$$

3(i) $f(x, y) = \dfrac{x^3 + y^3}{3}$**,** $x = 4\cos t$**,** $y = 3\sin t$

First compute partial derivatives:

$$f_x = x^2, \qquad f_y = y^2.$$

Compute the inner derivatives:

$$x'(t) = -4\sin t, \qquad y'(t) = 3\cos t.$$

Therefore

$$\frac{df}{dt} = x^2(-4\sin t) + y^2(3\cos t).$$

Substitute $x = 4\cos t$ and $y = 3\sin t$:

$$\frac{df}{dt} = (4\cos t)^2(-4\sin t) + (3\sin t)^2(3\cos t).$$

Thus

$$\frac{df}{dt} = -64\cos^2 t \sin t + 27\sin^2 t \cos t.$$

Answer. $\boxed{\dfrac{df}{dt} = -64\cos^2 t \sin t + 27\sin^2 t \cos t}$.

3(ii) $f(x, y) = x^2 + y^2 + xy$**,** $x = \sin t$**,** $y = e^t$

The partial derivatives are

$$f_x = 2x + y, \qquad f_y = 2y + x.$$

The inner derivatives are

$$x'(t) = \cos t, \qquad y'(t) = e^t.$$

Hence

$$\frac{df}{dt} = (2x + y)x'(t) + (2y + x)y'(t).$$

Substitute:

$$\frac{df}{dt} = (2\sin t + e^t)\cos t + (2e^t + \sin t)e^t.$$

Answer. $\boxed{\frac{df}{dt} = (2\sin t + e^t)\cos t + (2e^t + \sin t)e^t}$.

3(iii) $f(x, y) = \sqrt{1 + x^2 + y^2}$**,** $x = \ln t$**,** $y = \cos t$

The expression $\ln t$ requires $t > 0$. Compute partial derivatives:

$$f_x = \frac{x}{\sqrt{1 + x^2 + y^2}}, \qquad f_y = \frac{y}{\sqrt{1 + x^2 + y^2}}.$$

The inner derivatives are

$$x'(t) = \frac{1}{t}, \qquad y'(t) = -\sin t.$$

Therefore

$$\frac{df}{dt} = \frac{x}{\sqrt{1 + x^2 + y^2}} \cdot \frac{1}{t} + \frac{y}{\sqrt{1 + x^2 + y^2}}(-\sin t).$$

Substitute $x = \ln t$, $y = \cos t$:

$$\frac{df}{dt} = \frac{\frac{\ln t}{t} - \cos t \sin t}{\sqrt{1 + (\ln t)^2 + \cos^2 t}}.$$

Answer. $\boxed{\frac{df}{dt} = \frac{\frac{\ln t}{t} - \sin t \cos t}{\sqrt{1 + (\ln t)^2 + \cos^2 t}}}$, for $t > 0$.

3(iv) $f(x, y, z) = \ln \sqrt{x^2 + y^2 + z^2}$**,** $x = \sin t$**,** $y = \cos t$**,** $z = \tan t$

Write

$$f(x, y, z) = \frac{1}{2}\ln(x^2 + y^2 + z^2).$$

Therefore

$$f_x = \frac{x}{x^2 + y^2 + z^2}, \quad f_y = \frac{y}{x^2 + y^2 + z^2}, \quad f_z = \frac{z}{x^2 + y^2 + z^2}.$$

Also,

$$x'(t) = \cos t, \qquad y'(t) = -\sin t, \qquad z'(t) = \sec^2 t.$$

Thus

$$\frac{df}{dt} = \frac{x\cos t - y\sin t + z\sec^2 t}{x^2 + y^2 + z^2}.$$

Substitute $x = \sin t$, $y = \cos t$, $z = \tan t$. The numerator becomes

$$\sin t \cos t - \cos t \sin t + \tan t \sec^2 t = \tan t \sec^2 t.$$

The denominator is

$$\sin^2 t + \cos^2 t + \tan^2 t = 1 + \tan^2 t = \sec^2 t.$$

Therefore

$$\frac{df}{dt} = \frac{\tan t \sec^2 t}{\sec^2 t} = \tan t.$$

Answer. $\boxed{\frac{df}{dt} = \tan t}$, wherever all expressions are defined.

Problem 4: Implicit differentiation

Method. For one implicit equation $F(x, y) = 0$, if $F_y \neq 0$, then $\dfrac{dy}{dx} = -\dfrac{F_x}{F_y}$. For one implicit equation $F(x, y, z) = 0$ defining $z = z(x, y)$, if $F_z \neq 0$, then $\dfrac{\partial z}{\partial x} = -\dfrac{F_x}{F_z}$ and $\dfrac{\partial z}{\partial y} = -\dfrac{F_y}{F_z}$.

4(i)(a) $x^2 - 5xy + 2y^2 = 1$

Let

$$F(x, y) = x^2 - 5xy + 2y^2 - 1.$$

Then

$$F_x = 2x - 5y, \qquad F_y = -5x + 4y.$$

Therefore

$$\frac{dy}{dx} = -\frac{F_x}{F_y} = -\frac{2x - 5y}{-5x + 4y} = \frac{2x - 5y}{5x - 4y}.$$

Answer. $\boxed{\dfrac{dy}{dx} = \dfrac{2x - 5y}{5x - 4y}}$, where $5x - 4y \neq 0$.

4(i)(b) $x^3 + y^3 - 6xy = 0$

Let

$$F(x, y) = x^3 + y^3 - 6xy.$$

Then

$$F_x = 3x^2 - 6y, \qquad F_y = 3y^2 - 6x.$$

Hence

$$\frac{dy}{dx} = -\frac{3x^2 - 6y}{3y^2 - 6x} = \frac{2y - x^2}{y^2 - 2x}.$$

Answer. $\boxed{\dfrac{dy}{dx} = \dfrac{2y - x^2}{y^2 - 2x}}$, where $y^2 - 2x \neq 0$.

4(i)(c) $y\cos x = x^2$

Let

$$F(x,y) = y\cos x - x^2.$$

Then

$$F_x = -y\sin x - 2x, \qquad F_y = \cos x.$$

Therefore

$$\frac{dy}{dx} = -\frac{-y\sin x - 2x}{\cos x} = \frac{y\sin x + 2x}{\cos x}.$$

Answer. $\boxed{\dfrac{dy}{dx} = \dfrac{y\sin x + 2x}{\cos x}}$, where $\cos x \neq 0$.

4(i)(d) $y^2 = \dfrac{\cos(xy)}{1+\sin y}$

The printed problem is naturally read as the fraction form above. To avoid quotient algebra, first multiply both sides by $1+\sin y$:

$$y^2(1+\sin y) = \cos(xy).$$

Define

$$F(x,y) = y^2(1+\sin y) - \cos(xy).$$

Then

$$F_x = y\sin(xy),$$

because $\partial[-\cos(xy)]/\partial x = y\sin(xy)$. Also,

$$F_y = 2y(1+\sin y) + y^2\cos y + x\sin(xy).$$

Therefore

$$\frac{dy}{dx} = -\frac{F_x}{F_y} = -\frac{y\sin(xy)}{2y(1+\sin y) + y^2\cos y + x\sin(xy)}.$$

Answer. $\boxed{\dfrac{dy}{dx} = -\dfrac{y\sin(xy)}{2y(1+\sin y) + y^2\cos y + x\sin(xy)}}$.

4(ii)(a) $x^2 + 2y^2 + 3z^2 = 1$

Let

$$F(x,y,z) = x^2 + 2y^2 + 3z^2 - 1.$$

Then

$$F_x = 2x, \qquad F_y = 4y, \qquad F_z = 6z.$$

Hence

$$z_x = -\frac{2x}{6z} = -\frac{x}{3z}, \qquad z_y = -\frac{4y}{6z} = -\frac{2y}{3z}.$$

Answer. $\boxed{z_x = -\dfrac{x}{3z}}$, $\boxed{z_y = -\dfrac{2y}{3z}}$, where $z \neq 0$.

4(ii)(b) $x^2 - y^2 + z^2 - 2z = 4$

Let

$$F(x, y, z) = x^2 - y^2 + z^2 - 2z - 4.$$

Then

$$F_x = 2x, \qquad F_y = -2y, \qquad F_z = 2z - 2.$$

Thus

$$z_x = -\frac{2x}{2z-2} = -\frac{x}{z-1}, \qquad z_y = -\frac{-2y}{2z-2} = \frac{y}{z-1}.$$

Answer. $\boxed{z_x = -\dfrac{x}{z-1}}$, $\boxed{z_y = \dfrac{y}{z-1}}$, where $z \neq 1$.

4(ii)(c) $x^3 + y^3 + z^3 + 4xyz = 1$

Let

$$F(x, y, z) = x^3 + y^3 + z^3 + 4xyz - 1.$$

Then

$$F_x = 3x^2 + 4yz, \qquad F_y = 3y^2 + 4xz, \qquad F_z = 3z^2 + 4xy.$$

Therefore

$$z_x = -\frac{3x^2 + 4yz}{3z^2 + 4xy}, \qquad z_y = -\frac{3y^2 + 4xz}{3z^2 + 4xy}.$$

Answer. $\boxed{z_x = -\dfrac{3x^2 + 4yz}{3z^2 + 4xy}}$, $\boxed{z_y = -\dfrac{3y^2 + 4xz}{3z^2 + 4xy}}$.

4(ii)(d) $yz + x \ln y = z^2$

Let

$$F(x, y, z) = yz + x \ln y - z^2.$$

Then

$$F_x = \ln y, \qquad F_y = z + \frac{x}{y}, \qquad F_z = y - 2z.$$

Hence

$$z_x = -\frac{\ln y}{y - 2z}, \qquad z_y = -\frac{z + x/y}{y - 2z}.$$

Answer. $\boxed{z_x = -\dfrac{\ln y}{y - 2z}}$, $\boxed{z_y = -\dfrac{z + x/y}{y - 2z}}$, where $y > 0$ and $y \neq 2z$.

Problem 5: Mean value theorem

5(i) Convexity of $S = \{x : \|x - x_0\| < r\} \subset \mathbb{R}^n$

Method. To prove a set is convex, take any two points in the set and show that every point on the straight segment joining them is still in the set.

Let $a, b \in S$ and let $\lambda \in [0, 1]$. We must show that

$$\lambda a + (1 - \lambda)b \in S.$$

By definition of S,

$$\|a - x_0\| < r, \qquad \|b - x_0\| < r.$$

Now compute the distance from the convex combination to x_0:

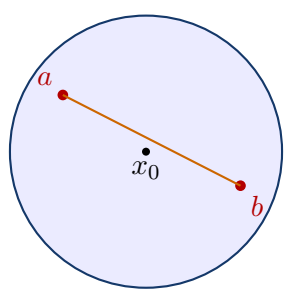
$$\lambda a + (1 - \lambda)b - x_0 = \lambda(a - x_0) + (1 - \lambda)(b - x_0).$$

Using the triangle inequality and homogeneity of the norm,

$$\begin{aligned}
\|\lambda a + (1 - \lambda)b - x_0\| &= \|\lambda(a - x_0) + (1 - \lambda)(b - x_0)\| \\
&\leq \lambda \|a - x_0\| + (1 - \lambda) \|b - x_0\| \\
&< \lambda r + (1 - \lambda)r \\
&= r.
\end{aligned}$$

Therefore $\lambda a + (1 - \lambda)b \in S$.

Answer. $\boxed{S \text{ is convex.}}$



Every line segment between two points in an open ball stays inside the ball.

5(ii) If $D_u f(x) = 0$ in every unit direction on an open convex set, then f is constant

Let $a, b \in S$. If $a = b$, then clearly $f(a) = f(b)$. Assume $a \neq b$. Define

$$u = \frac{b - a}{\|b - a\|}.$$

Then u is a unit vector. Because S is convex, every point of the segment

$$\gamma(t) = a + t(b - a), \qquad 0 \leq t \leq 1,$$

lies in S. Define a one-variable function

$$g(t) = f(\gamma(t)) = f(a + t(b - a)).$$

The derivative of g is the rate of change of f along the direction u, scaled by the speed $\|b - a\|$:

$$g'(t) = \|b - a\| \; D_u f(\gamma(t)).$$

By hypothesis, $D_u f(x) = 0$ for every $x \in S$ and every unit vector u. Therefore

$$g'(t) = 0$$

for every $t \in [0, 1]$ where the derivative is taken. By the one-variable mean value theorem, g is constant on $[0, 1]$. Hence

$$g(0) = g(1).$$

But

$$g(0) = f(a), \qquad g(1) = f(b).$$

Therefore

$$f(a) = f(b).$$

Since $a, b \in S$ were arbitrary, f is constant on S.

Answer. $\boxed{f \text{ is constant on } S.}$

5(iii) For differentiable f on open convex S, prove $\nabla f = 0$ iff f is constant

First suppose f is constant on S. Then for each coordinate direction e_i,

$$\frac{\partial f}{\partial x_i}(x) = 0$$

for all $x \in S$. Therefore

$$\nabla f(x) = 0$$

for all $x \in S$.

Conversely, suppose

$$\nabla f(x) = 0$$

for all $x \in S$. For any unit vector u,

$$D_u f(x) = \nabla f(x) \cdot u = 0 \cdot u = 0.$$

By part (ii), f is constant on S.

Answer. $\boxed{\nabla f = 0 \text{ on } S \iff f \text{ is constant on } S.}$

Problem 6: Linear maps

Method. A linear map is determined by what it does to basis vectors. In coordinates, every linear map is represented by a matrix. Geometric transformations such as rotations, reflections, and projections are therefore written as matrices acting on coordinate vectors.

6(i) Rotation in $\mathbb{R}^2$ anticlockwise by angle θ

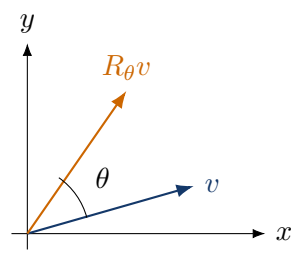
The standard anticlockwise rotation matrix is

$$R_\theta = \begin{bmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{bmatrix}.$$

Therefore

$$R_\theta \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} x\cos\theta - y\sin\theta \\ x\sin\theta + y\cos\theta \end{bmatrix}.$$

Answer. $\boxed{R_\theta = \begin{bmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{bmatrix}}$.



6(ii) Reflection about a line making angle θ with the x-axis

Reflection across the x-axis is

$$\begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}.$$

To reflect across a line making angle θ, rotate the line to the x-axis, reflect, then rotate back:

$$M_\theta = R_\theta \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix} R_{-\theta}.$$

Carrying out the multiplication gives

$$M_\theta = \begin{bmatrix} \cos 2\theta & \sin 2\theta \\ \sin 2\theta & -\cos 2\theta \end{bmatrix}.$$

Answer. $\boxed{M_\theta = \begin{bmatrix} \cos 2\theta & \sin 2\theta \\ \sin 2\theta & -\cos 2\theta \end{bmatrix}}$.

6(iii) Projection onto a line through the origin

Let the line be spanned by a nonzero vector $v \in \mathbb{R}^n$. The projection of x onto the line is the component of x in the direction of v:

$$\text{proj}_v(x) = \frac{x \cdot v}{v \cdot v} v.$$

If u is the unit vector in the direction of the line, then $v \cdot v = 1$ and

$$\text{proj}_u(x) = (x \cdot u)u.$$

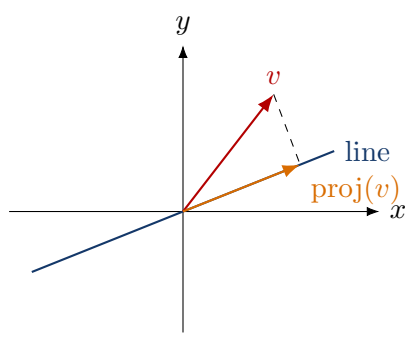
In matrix form,

$$P = \frac{vv^T}{v^T v}.$$

For a line in $\mathbb{R}^2$ making angle θ with the x-axis, take $u = (\cos\theta, \sin\theta)$. Then

$$P_\theta = uu^T = \begin{bmatrix} \cos^2\theta & \sin\theta\cos\theta \\ \sin\theta\cos\theta & \sin^2\theta \end{bmatrix}.$$

Answer. $\boxed{\text{proj}_v(x) = \frac{x \cdot v}{v \cdot v} v}$ and $\boxed{P = \frac{vv^T}{v^T v}}$.



Projection drops the vector perpendicularly onto the line.

6(iv) Map from $\mathbb{R}^3$ onto the xy-plane

The xy-plane consists of all points whose third coordinate is 0. Therefore the projection map is

$$P(x, y, z) = (x, y, 0).$$

In matrix form,

$$P = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 0 \end{bmatrix}.$$

Answer. $\boxed{P(x, y, z) = (x, y, 0)}$ and $\boxed{P = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 0 \end{bmatrix}}$.

6(v) Every linear map $A : \mathbb{R}^n \to \mathbb{R}^m$ has an $m \times n$ matrix

Let $e_1, \ldots, e_n$ be the standard basis of $\mathbb{R}^n$. Every vector $x \in \mathbb{R}^n$ can be written as

$$x = x_1 e_1 + x_2 e_2 + \cdots + x_n e_n.$$

Since A is linear,

$$A(x) = x_1 A(e_1) + x_2 A(e_2) + \cdots + x_n A(e_n).$$

Now form the matrix $\widetilde{A}$ whose jth column is $A(e_j)$:

$$\widetilde{A} = \begin{bmatrix} A(e_1) & A(e_2) & \cdots & A(e_n) \end{bmatrix}.$$

Then matrix multiplication gives exactly

$$A(x) = \widetilde{A} \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{bmatrix}.$$

Conversely, any $m \times n$ matrix defines a linear map by multiplication because matrix multiplication preserves sums and scalar multiples.

Answer. $\boxed{\text{Linear maps } \mathbb{R}^n \to \mathbb{R}^m \text{ are exactly } m \times n \text{ matrices.}}$

Problems 7 and 8: Second-degree Taylor polynomials and Hessians at critical points

Method. First solve $\nabla f = 0$ to find the critical points. Then compute the Hessian matrix

$$H_f(a,b) = \begin{bmatrix} f_{xx}(a,b) & f_{xy}(a,b) \\ f_{yx}(a,b) & f_{yy}(a,b) \end{bmatrix}.$$

At a critical point (a,b), the second-degree Taylor polynomial is

$$T_2(x,y) = f(a,b) + \frac{1}{2}\begin{bmatrix} x-a & y-b \end{bmatrix} H_f(a,b) \begin{bmatrix} x-a \\ y-b \end{bmatrix},$$

because the first-order term is zero when $\nabla f(a,b) = 0$.

$$\boxed{\text{Solve } \nabla f = 0} \longrightarrow \boxed{\text{Compute Hessian } H_f} \longrightarrow \boxed{\text{Insert into } T_2}$$

7(i), 8(i) $f(x,y) = xy^2 + x^2y$

Compute the gradient:

$$f_x = y^2 + 2xy = y(y+2x), \qquad f_y = 2xy + x^2 = x(2y+x).$$

Set both equal to zero:

$$y(y+2x) = 0, \qquad x(2y+x) = 0.$$

Checking the cases gives only

$$(x,y) = (0,0).$$

The Hessian entries are

$$f_{xx} = 2y, \qquad f_{xy} = 2x + 2y, \qquad f_{yy} = 2x.$$

At $(0,0)$,

$$H_f(0,0) = \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix}.$$

Also $f(0,0) = 0$. Therefore

$$T_2(x,y) = 0.$$

Answer. Critical point: $\boxed{(0,0)}$. Hessian: $\boxed{\begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix}}$. Taylor polynomial: $\boxed{T_2 = 0}$.

7(ii), 8(ii) $f(x,y) = x^2 + (y-1)^2$

$$f_x = 2x, \qquad f_y = 2(y-1).$$

Thus $\nabla f = 0$ gives

$$x = 0, \qquad y = 1.$$

So the critical point is $(0,1)$. The Hessian is constant:

$$H_f = \begin{bmatrix} 2 & 0 \\ 0 & 2 \end{bmatrix}.$$

Since $f(0,1) = 0$,

$$T_2(x,y) = x^2 + (y-1)^2.$$

Answer. Critical point: $\boxed{(0,1)}$. Hessian: $\boxed{\begin{bmatrix} 2 & 0 \\ 0 & 2 \end{bmatrix}}$. Taylor polynomial: $\boxed{T_2 = x^2 + (y-1)^2}$.

7(iii), 8(iii) $f(x,y) = 1 + x^2 - (y-1)^2$

$$f_x = 2x, \qquad f_y = -2(y-1).$$

Thus $\nabla f = 0$ gives $(0,1)$. The Hessian is

$$H_f = \begin{bmatrix} 2 & 0 \\ 0 & -2 \end{bmatrix}.$$

Since $f(0,1) = 1$,

$$T_2(x,y) = 1 + x^2 - (y-1)^2.$$

Answer. Critical point: $\boxed{(0,1)}$. Hessian: $\boxed{\begin{bmatrix} 2 & 0 \\ 0 & -2 \end{bmatrix}}$. Taylor polynomial: $\boxed{T_2 = 1 + x^2 - (y-1)^2}$.

7(iv), 8(iv) $f(x,y) = (x - y + 1)^2$

Let $q = x - y + 1$. Then

$$f_x = 2q, \qquad f_y = -2q.$$

Thus $\nabla f = 0$ exactly when

$$q = 0, \qquad \text{that is} \qquad y = x + 1.$$

So every point $(a, a+1)$ is critical. The Hessian is constant:

$$H_f = \begin{bmatrix} 2 & -2 \\ -2 & 2 \end{bmatrix}.$$

At a critical point $(a, a+1)$, $f(a, a+1) = 0$. Let

$$u = x - a, \qquad v = y - (a+1).$$

Then

$$T_2(x, y) = \frac{1}{2}\begin{bmatrix} u & v \end{bmatrix}\begin{bmatrix} 2 & -2 \\ -2 & 2 \end{bmatrix}\begin{bmatrix} u \\ v \end{bmatrix} = (u - v)^2.$$

Since $u - v = (x - a) - (y - a - 1) = x - y + 1$,

$$T_2(x, y) = (x - y + 1)^2.$$

Answer. Critical set: $\boxed{y = x + 1}$. Hessian: $\boxed{\begin{bmatrix} 2 & -2 \\ -2 & 2 \end{bmatrix}}$. Taylor polynomial at $(a, a+1)$: $\boxed{T_2 = (x - y + 1)^2}$.

7(v), 8(v) $f(x, y) = \sin x \sin y \sin(x + y)$ on $[0, \pi]^2$

First compute the first partial derivatives:

$$\begin{aligned} f_x &= \cos x \sin y \sin(x + y) + \sin x \sin y \cos(x + y) \\ &= \sin y \sin(2x + y), \end{aligned}$$

and similarly

$$f_y = \sin x \sin(x + 2y).$$

Thus $\nabla f = 0$ means

$$\sin y \sin(2x + y) = 0, \qquad \sin x \sin(x + 2y) = 0.$$

On the boundary, this gives the four corners

$$(0, 0), \quad (\pi, 0), \quad (0, \pi), \quad (\pi, \pi).$$

In the interior, $\sin x \neq 0$ and $\sin y \neq 0$, so

$$\sin(2x + y) = 0, \qquad \sin(x + 2y) = 0.$$

Since $x, y \in (0, \pi)$, the only interior solutions are

$$(x, y) = \left(\frac{\pi}{3}, \frac{\pi}{3}\right), \qquad (x, y) = \left(\frac{2\pi}{3}, \frac{2\pi}{3}\right).$$

Now compute the second derivatives:

$$f_{xx} = 2 \sin y \cos(2x + y), \qquad f_{xy} = \sin(2x + 2y), \qquad f_{yy} = 2 \sin x \cos(x + 2y).$$

At the four corners, $H_f = 0$ and $f = 0$, so $T_2 = 0$.

At $p_1 = (\pi/3, \pi/3)$,

$$f(p_1) = \frac{3\sqrt{3}}{8}, \qquad H_f(p_1) = \begin{bmatrix} -\sqrt{3} & -\sqrt{3}/2 \\ -\sqrt{3}/2 & -\sqrt{3} \end{bmatrix}.$$

Let $u = x - \pi/3$ and $v = y - \pi/3$. Then

$$T_2 = \frac{3\sqrt{3}}{8} - \frac{\sqrt{3}}{2}u^2 - \frac{\sqrt{3}}{2}uv - \frac{\sqrt{3}}{2}v^2.$$

At $p_2 = (2\pi/3, 2\pi/3)$,

$$f(p_2) = -\frac{3\sqrt{3}}{8}, \qquad H_f(p_2) = \begin{bmatrix} \sqrt{3} & \sqrt{3}/2 \\ \sqrt{3}/2 & \sqrt{3} \end{bmatrix}.$$

Let $u = x - 2\pi/3$ and $v = y - 2\pi/3$. Then

$$T_2 = -\frac{3\sqrt{3}}{8} + \frac{\sqrt{3}}{2}u^2 + \frac{\sqrt{3}}{2}uv + \frac{\sqrt{3}}{2}v^2.$$

Answer. Critical points are $\boxed{(0,0), (\pi,0), (0,\pi), (\pi,\pi), (\pi/3,\pi/3), (2\pi/3,2\pi/3)}$. The Hessians and Taylor polynomials are listed above explicitly.

7(vi), 8(vi) $f(x,y) = x^2 - xy + y^2 - 2x + y$

$$f_x = 2x - y - 2, \qquad f_y = -x + 2y + 1.$$

Set both equal to zero:

$$2x - y - 2 = 0, \qquad -x + 2y + 1 = 0.$$

From the first equation $y = 2x - 2$. Substitute into the second:

$$-x + 2(2x - 2) + 1 = 0 \quad \Longrightarrow \quad 3x - 3 = 0.$$

Thus $x = 1$ and $y = 0$. The Hessian is constant:

$$H_f = \begin{bmatrix} 2 & -1 \\ -1 & 2 \end{bmatrix}.$$

Also $f(1,0) = -1$. Let $u = x - 1$ and $v = y$. Then

$$T_2 = -1 + u^2 - uv + v^2.$$

Answer. Critical point: $\boxed{(1,0)}$. Hessian: $\boxed{\begin{bmatrix} 2 & -1 \\ -1 & 2 \end{bmatrix}}$. Taylor polynomial: $\boxed{T_2 = -1 + (x-1)^2 - (x-1)y + y^2}$.

7(vii), 8(vii) $f(x,y) = 2x^2 - xy - 3y^2 - 3x + 7y$

$$f_x = 4x - y - 3, \qquad f_y = -x - 6y + 7.$$

Set both equal to zero:

$$4x - y - 3 = 0, \qquad -x - 6y + 7 = 0.$$

The first gives $y = 4x - 3$. Substitute:

$$-x - 6(4x - 3) + 7 = 0 \quad \Longrightarrow \quad -25x + 25 = 0.$$

So $x = 1$, and then $y = 1$. The Hessian is

$$H_f = \begin{bmatrix} 4 & -1 \\ -1 & -6 \end{bmatrix}.$$

Also $f(1,1) = 2$. Let $u = x - 1$, $v = y - 1$. Then

$$T_2 = 2 + 2u^2 - uv - 3v^2.$$

Answer. Critical point: $\boxed{(1,1)}$. Hessian: $\boxed{\begin{bmatrix} 4 & -1 \\ -1 & -6 \end{bmatrix}}$. Taylor polynomial: $\boxed{T_2 = 2 + 2(x-1)^2 - (x-1)(y-1) - 3(y-1)^2}$.

7(viii), 8(viii) $f(x,y) = x^3 - 3xy^2 + y^3$

$$f_x = 3x^2 - 3y^2 = 3(x-y)(x+y), \qquad f_y = -6xy + 3y^2 = 3y(y - 2x).$$

Set both equal to zero. From $f_y = 0$, either $y = 0$ or $y = 2x$. If $y = 0$, then $f_x = 3x^2$, so $x = 0$. If $y = 2x$, then

$$f_x = 3x^2 - 3(2x)^2 = -9x^2,$$

so again $x = 0$ and $y = 0$. Thus the only critical point is $(0,0)$.

The second derivatives are

$$f_{xx} = 6x, \qquad f_{xy} = -6y, \qquad f_{yy} = -6x + 6y.$$

At $(0,0)$,

$$H_f(0,0) = \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix}.$$

Since $f(0,0) = 0$,

$$T_2 = 0.$$

Answer. Critical point: $\boxed{(0,0)}$. Hessian: $\boxed{0}$. Taylor polynomial: $\boxed{T_2 = 0}$.

7(ix), 8(ix) $f(x,y) = x^2y^3(6-x-y)$

First compute

$$f_x = xy^3(12-3x-2y), \qquad f_y = x^2y^2(18-3x-4y).$$

Thus $\nabla f = 0$ if either $x = 0$, or $y = 0$, or both linear factors vanish:

$$12-3x-2y = 0, \qquad 18-3x-4y = 0.$$

Solving these two equations gives

$$y = 3, \qquad x = 2.$$

Therefore the critical set is

$$\{(0,b) : b \in \mathbb{R}\} \cup \{(a,0) : a \in \mathbb{R}\} \cup \{(2,3)\}.$$

The second derivatives are

$$\begin{aligned} f_{xx} &= y^3(12-6x-2y), \\ f_{xy} &= xy^2(36-9x-8y), \\ f_{yy} &= x^2y(36-6x-12y). \end{aligned}$$

At $(0,b)$,

$$H_f(0,b) = \begin{bmatrix} b^3(12-2b) & 0 \\ 0 & 0 \end{bmatrix}.$$

Since $f(0,b) = 0$, the Taylor polynomial at $(0,b)$ is

$$T_2 = b^3(6-b)x^2.$$

At $(a,0)$,

$$H_f(a,0) = \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix}, \qquad T_2 = 0.$$

At $(2,3)$,

$$f(2,3) = 2^2 \cdot 3^3(6-2-3) = 108,$$

and

$$H_f(2,3) = \begin{bmatrix} -162 & -108 \\ -108 & -144 \end{bmatrix}.$$

Let $u = x-2$, $v = y-3$. Then

$$T_2 = 108 - 81u^2 - 108uv - 72v^2.$$

Answer. Critical set: $\boxed{x = 0 \text{ or } y = 0 \text{ or } (x,y) = (2,3)}$. Hessians and Taylor polynomials are as computed above.

7(x), 8(x) $f(x,y) = e^{2x+3y}(8x^2 - 6xy + 3y^2)$

Let

$$q(x,y) = 8x^2 - 6xy + 3y^2, \qquad f = e^{2x+3y}q.$$

Then

$$f_x = e^{2x+3y}(2q + q_x), \qquad f_y = e^{2x+3y}(3q + q_y),$$

where

$$q_x = 16x - 6y, \qquad q_y = -6x + 6y.$$

Because $e^{2x+3y} \neq 0$, the critical point equations are

$$2q + 16x - 6y = 0, \qquad 3q - 6x + 6y = 0.$$

From the second equation,

$$q = 2x - 2y.$$

From the first equation,

$$q = -8x + 3y.$$

Equating these two expressions for q gives

$$2x - 2y = -8x + 3y \quad \Longrightarrow \quad 10x = 5y \quad \Longrightarrow \quad y = 2x.$$

Substitute $y = 2x$ into $q = 2x - 2y$:

$$q = 2x - 4x = -2x.$$

But directly from $q = 8x^2 - 6x(2x) + 3(2x)^2$,

$$q = 8x^2 - 12x^2 + 12x^2 = 8x^2.$$

Thus

$$8x^2 = -2x \quad \Longrightarrow \quad 2x(4x+1) = 0.$$

Hence

$$x = 0 \quad \text{or} \quad x = -\frac{1}{4}.$$

Using $y = 2x$, the critical points are

$$(0,0), \qquad \left(-\frac{1}{4}, -\frac{1}{2}\right).$$

At a critical point the Hessian simplifies to the derivative of the bracketed first-derivative factors. The resulting Hessian values are

$$H_f(0,0) = \begin{bmatrix} 16 & -6 \\ -6 & 6 \end{bmatrix},$$

and

$$H_f\left(-\frac{1}{4}, -\frac{1}{2}\right) = e^{-2}\begin{bmatrix} 14 & -9 \\ -9 & 3/2 \end{bmatrix}.$$

At $(0,0)$, $f(0,0) = 0$, so

$$T_2 = 8x^2 - 6xy + 3y^2.$$

At $p = (-1/4, -1/2)$,

$$f(p) = e^{-2} \cdot \frac{1}{2} = \frac{e^{-2}}{2}.$$

Let

$$u = x + \frac{1}{4}, \qquad v = y + \frac{1}{2}.$$

Then

$$T_2 = \frac{e^{-2}}{2} + e^{-2}\left(7u^2 - 9uv + \frac{3}{4}v^2\right).$$

Answer. Critical points: $\boxed{(0,0)}$ and $\boxed{(-1/4, -1/2)}$. Hessians and Taylor polynomials are as above.

Problem 9: First and second degree Taylor polynomial of $f(x, y) = e^x \cos y$ at the origin

Method. For Taylor expansion at $(0, 0)$, compute $f(0, 0)$, the gradient at $(0, 0)$, and the Hessian at $(0, 0)$. Then use

$$T_1 = f(0,0) + f_x(0,0)x + f_y(0,0)y,$$

and

$$T_2 = T_1 + \frac{1}{2}\left(f_{xx}(0,0)x^2 + 2f_{xy}(0,0)xy + f_{yy}(0,0)y^2\right).$$

First,

$$f(0,0) = e^0 \cos 0 = 1.$$

The first partial derivatives are

$$f_x = e^x \cos y, \qquad f_y = -e^x \sin y.$$

At $(0, 0)$,

$$f_x(0,0) = 1, \qquad f_y(0,0) = 0.$$

Therefore the first-degree Taylor polynomial is

$$T_1(x, y) = 1 + x.$$

Now compute second partial derivatives:

$$f_{xx} = e^x \cos y, \qquad f_{xy} = -e^x \sin y, \qquad f_{yy} = -e^x \cos y.$$

At $(0, 0)$,

$$f_{xx}(0,0) = 1, \qquad f_{xy}(0,0) = 0, \qquad f_{yy}(0,0) = -1.$$

Thus

$$T_2(x, y) = 1 + x + \frac{1}{2}(x^2 - y^2).$$

Answer. $\boxed{T_1(x, y) = 1 + x}$ and $\boxed{T_2(x, y) = 1 + x + \frac{1}{2}(x^2 - y^2)}$.

Final checklist of core formulas

Topic	Formula used
Level curve tangent	$F_x(P)(x - x_0) + F_y(P)(y - y_0) = 0$
Level surface tangent plane	$\nabla F(P) \cdot ((x, y, z) - P) = 0$
Chain rule	$\frac{df}{dt} = \nabla f \cdot \frac{d}{dt}(x, y, z)$

Implicit derivative	$\dfrac{dy}{dx} = -F_x/F_y$
Implicit partials	$z_x = -F_x/F_z$, $z_y = -F_y/F_z$
Gradient-zero theorem	On an open convex set, differentiable f is constant iff $\nabla f = 0$
Taylor at critical point	$T_2 = f(P) + \frac{1}{2}(h^T H_f(P)h)$